FYMCA

SMCA511
Implementation of Programming Concepts

PRACITCAL LAB ASSESSMENTS

Assignment Completion & Viva Sheet

Name: _____

Roll No: _____

	Programming Concepts		
Sr. No	**Assignment Name**	**Marks (out of 5)**	**Signature**
1.	Use of data types, simple operators (expressions)		
2.	Use of decision making statements (if and if-else, nested structures, switch case)		
3.	Use of simple loops, nested loops		
4.	Use of standard library functions and menu driven programs		
5.	Use of user defined functions, recursive functions		
6.	Use of arrays (1-d arrays),multidimensional array(2-d arrays) and functions		
7.	Use of Simple Pointers		
8.	Dynamic Memory Allocation		
9.	String Handling Using Standard Library Functions.		
10.	String Handling Using User defined Functions.		
11.	Structure and Unions.		
12.	File Handling (Text Files).		
13.	Command Line Arguments and Pre-processor Directives.		
Total Marks (out of 65)			
Total Marks(out of 15) (Assignment(10) +Viva(5))			

Name & Sign of Batch In-charge **Head of School**

Date:

Assignment 1

Set A . Apply all the three program development steps for the following examples.

1. Write a program to accept an integer n and display all even numbers upto n.
2. Accept two integers x and y and calculate the sum of all integers between x and y (both inclusive)
3. Write a program to accept two integers x and n and compute x^n
4. Write a program to accept an integer and check if it is prime or not.
5. Write a program to accept an integer and count the number of digits in the number.
6. Write a program to accept a character, an integer n and display the next n characters.
7. Write a program to display all prime numbers between___and ___
8. Write a program to display multiplication tables from to having n multiples each. The output should be displayed in a tabular format. For example, the multiplication tables of 2 to 9 having 10 multiples each is shown below.

   2 □ 1 = 2 3 □ 1 = 3 ……9 □ 1 = 9

   2 □ 2 = 4 3 □ 2 = 6……..9 □ 2 = 18

   ……. …….

   2 □ 10 = 20 3 □ 10 = 30……..9 □ 10 = 90

9. Modify the sample program 1 to display n lines as follows (here n=4).

A	B	C	D
E	F	G	
H	I		
J			

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Set B. Apply all the three program development steps for the following examples.

1. Write a program to accept real number x and integer n and calculate the sum of first n terms of the series x+ 3x+5x+7x+…
2. Write a program to accept real number x and integer n and calculate the sum of first n terms of the series

$$\text{Sum} = \frac{1}{x} + \frac{2}{x^2} + \frac{3}{x^3} + \cdots + \frac{n}{x^n}$$

Set C. Write C programs to solve the following problems

1. Write a program which accepts a number and checks if the number is a palindrome (Hint number = reverse of number)
   Example: number = 3472 Output: It is not a palindrome, number = 262, Output : It is a palindrome
2. Write a program to display all Armstrong numbers between 1 and 500. (An Armstrong number is a number such that the sum of cube of digits = number itself Ex. 153 = 1*1*1 + 5*5*5 + 3*3*3
3. Display all perfect numbers below 500. [A perfect number is a number, such that the sum of its factors is equal to the number itself]. Example: 6 (1 + 2 + 3), 28 (1+2+4+7+14)

Assignment Evaluation

0: Not Done [ ]	1: Incomplete [ ]	2: Late Submission [ ]
3: Needs Improvement [ ]	4: Complete [ ]	5: Well Done [ ]

Signature of Instructor **Date**

Assignment 2

Set A . Write C programs for the following problems

1. Write a menu driven program to perform the following operations till the user selects Exit. Accept appropriate data for each option. Use standard library functions from math.h
   i. Sine ii. Cosine iii. log iv. e^x v. Square Root vi. Exit
2. Accept two complex numbers from the user (real part, imaginary part). Write a menut driven program to perform the following operations till the user selects Exit.
   i. ADD ii. SUBTRACT iii. MULTIPLY iv. EXIT

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Set B. Write C programs for the following problems

1. Write a menu driven program to perform the following operations on a character type variable.
   i. Check if it is an alphabet
   ii. Check if it is a digit. If it is a digit find its cube (Use math.h)
   iii. Check if it is a lowercase alphabet. Display its next character in uppercase.
   iv. Check if it is a uppercase alphabet. Display its next character in lowercase.
   v. Exit

Assignment Evaluation

0: Not Done [ ]	1: Incomplete [ ]	2: Late Submission [ ]
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Signature of Instructor	**Date**

Assignment 3

Set A . Write C programs for the following problems

1. Write a function Max, which accept two integers as parameters and returns maximum number to the calling function. Use this function in main to accept 3 numbers.
2. Write a function isPrime, which accepts an integer as parameter and returns 1 if the number is prime, 0 otherwise. Use this function in main to accept n numbers and check if they are prime or not. (Till user Quit)
3.

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Set B . Write C programs for the following problems

1. Write a function isEven, which accepts an integer as parameter and returns 1 if the number is even, and 0 otherwise. Use this function in main to accept n numbers and check if they are even or odd.
2. Write a function, which accepts a character and integer n as parameter and displays the next n characters.

Set C . Write C programs for the following problems

1. Write a recursive function to calculate the nth Fibonacci number. Use this function in main to display the first n Fibonacci numbers. The recursive definition of nth Fibonacci number is as follows:

$$fib(n)=1 \quad \text{if } n = 1 \text{ or } 2$$
$$= fib(n-2) + fib(n-1) \quad \text{if } n>2$$

2. Write a function that accepts a character as parameter and returns 1 if it is an alphabet, 2 if it is a digit and 3 if it is a special symbol. In main, accept characters till the user enters EOF and use the function to count the total number of alphabets, digits and special symbols entered.

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Assignment 4

Set A

1. Write a program to accept n numbers in an array and display the largest and smallest number of the array.
2. Write a program to accept numbers in the range of 1 to 25. Enter -1 to stop accepting the numbers from the user. Further accept a digit from the user from the given range and find the number of times that number has been entered (i.e. Hit Count of that number).
3. Write a function, which accepts an integer array and an integer as parameters and count the occurrences of the number in that array.

   Example:

   Input 1 5 2 1 6 3 8 2 9 15 1 30

   Number to search : 1 Output: 1 occurs 3 times
4. Write a program to add two matrices. Write separate functions to accept, add and display the matrices.

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Set B

1. Write a program to accept n numbers in an array and calculate the average of n numbers.
2. Write a program to find the sum of diagonal elements of a matrix.

Set C

1. Write a program to remove all duplicate elements from an array.
2. Write a program to accept an mXn matrix and display an m+1 X n+1 matrix such that the m+1th row contains the sum of all elements of corresponding row and the n+1th column contains the sum of elements of the corresponding column.

Example:

A			B			
1	2	3	1	2	3	6
4	5	6	4	5	6	15
7	8	9	7	8	9	24
			12	15	18	45

Assignment Evaluation

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Assignment 5

Set A: Write C programs for the following problems.

1. Write a program to read two integers using pointers and perform all arithmetic operations on them.
2. Write a program to accept an integer using pointer and check whether it is even or odd.
3. Write a program to find maximum from two integers using pointers.
4. Write a program to display the elements of an array containing n integers in the reverse order using a pointer to the array.

Signature of the instructor [ ] Date [/ /]

Set B. Write C programs for the following problems.

1. Write a program to read N integers in an array using pointers and display minimum from them.

   (Hint: Pass the addresses of the variables to the function to get area and perimeter)
2. Accept n integers in array A in main. Write a function which takes this array as parameter and find minimum and maximum from it. Display these values in main.

Set C. Write programs to solve the following problems

1. Accept date (dd, mm yy). Write a function to add no of days to the date and display the new date. Pass dd, mm and yy to the function using pointers.
2. Write a function which accepts a number and three flags as parameters. If the number is even, set the first flagto1.If the number is prime, set the second flag to1.If the number is divisible by 3 or 7, set the third flag to 1. In main, accept an integer and use this function to check if it is even, prime and divisible by 3 or 7. (Hint : pass the addresses of flags to the function)

Assignment Evaluation

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Assignment 6

Set A. Write C programs for the following problems.

1. Write a C program to dynamically allocate memory for an array of ten integers using malloc, accept five integers from the user, print all the entered integers,
2. Write a program to accept N integers and store them dynamically display them in reverse order.
3. Write a program to allocate memory dynamically for n integers such that the memory is initialized to 0. And display it.

Signature of the instructor [ ] Date [/ /]

Set B. Write C programs for the following problems.

1. Accept N integers in an array using dynamic memory allocation. Find maximum from them and display.
2. Accept n integers in an array. Copy only the non-zero elements to another array (Using dynamic memory allocation). Calculate the sum and average of non-zero elements.
3. Accept n integers in an array. Calculate the sum of array elements (Using dynamic memory allocation).

Set C. Write programs to solve the following problems

1. Accept a matrix of order mXn (Use dynamic memory allocation and array of pointers concept). Display Column wise sum of elements.

Assignment Evaluation

0: Not Done [ ]	1: Incomplete [ ]	2: Late Submission [ ]
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Assignment 7

Set A. Write C programs for the following problems.

1. Write a program to read a string and copy it to another sting and display copied string and the length of copied string. (Use built in functions)
2. Write a program to read two strings. If first string is greater than second then concatenate second to first and display concatenated string, If first string is smaller than second then concatenate first to second, other-wise display length of string. (use strcmp)
3. Write a program to read a string and one character. Check whether given character is present in the given string or not? (Hint: use strchr or strrchr)

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Set B. Write C programs for the following problems.

1. Write a program which accepts a sentence from the user and alters it as follows: Every space is replaced by *, case of all alphabets is reversed, digits are replaced by ?.
2. Write a program which accepts a string from the user. Find and display reverse of it. (Don't use any function).
3. Write a menu driven program to perform the following operations on strings using standard library functions:
   1. Length 2. Copy 3.Concatenation 4. Compare

Assignment Evaluation

0: Not Done [ ]	1: Incomplete [ ]	2: Late Submission [ ]
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Assignment 8

Set A. Write C programs for the following problems.

1. Write a program to accept a string and find its length using user defined function. (Don't use library functions)
2. Write a program to find the number of vowels in a string.
3. Write a function that takes a string as parameter and returns the same string in uppercase (use pointes). Accept this string in main and display converted string in main only.

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Set B. Write C programs for the following problems.

1. Write a program to compare two strings using pointers(Don't use library functions).
2. Write a program to accept a string and check whether string is palindrome or not.
3. Write a program that will accept a string and character to search. The program will call a function, which will find occurrence of given character in the string. Display this count in main() function.

Set C. Write programs to solve the following problems

1. Write a Menu driven program to perform following operations on strings till user selects exit.
   1. Copy 2. Concatenate 3. Length 4. Reverse
2. Write a program to count frequency of each character in a string.

   For Example:-If input string is "This is Monitor"

   then output should be Character t occurs 2 times
   Character h occurs 1 times
   Character i occurs 3 times
   Character s occurs 2 times
   Character m occurs 1 times
   Character o occurs 2 times
   Character n occurs 1 times
   Character r occurs 1 times.

Assignment Evaluation

0: Not Done [ ]	1: Incomplete [ ]	2: Late Submission [ ]
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Signature of Instructor	**Date**

Assignment 9

Set A. Write C programs for the following problems.

1. Create a structure employee (id, name, salary). Accept details of n employees and display it in summary format.
2. Create a structure employee(id, name, joining date(dd, mm, yyyy)). Accept the details for n employees. Accept month and display the details of employee having birthdate in that month. (Hint : Use Nested structure)

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Set B. Write C programs for the following problems.

1. Create a structure item(item number, item name, rate, qty, total). Accept details of n items (calculate total as qty * rate). And display Bill in the following format

Sr. No.	Item Name	Rate	Qty	Total
1	Pen	10	3	30.00
2	Eraser	5	1	05.00
			Grand Total :	35.00

2. Create a structure Book detail and display the book details in proper format by passing the structure as function argument.

Set C. Write programs to solve the following problems

1. Write a program to declare structure employee having data member name, street, city. Accept data for three employee and display it. (Hint: Array of structure)
2. Create a structure employee (id, name, joining date(dd, mm, yyyy)). Accept the details for n employees. Accept month and display the details of employee having birthdate in that month. (Hint : Use Nested structure)

Assignment Evaluation

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Signature of Instructor **Date**

Assignment 10

Set A. Write C programs for the following problems.

1. Write a program to accept a file name and a single character. Display the count indicating total number of times character occurs in the file
2. A text file contains lines of text. Write a program which removes all extra spaces from the file.
3. Write a program to copy contents of one file to another file.
4. Write a program to accept a filename and count the number of words, lines and characters in the file.

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Set B. Write C programs for the following problems.

1. Write a program to read integers from a file and write even and odd numbers in separate files.
2. Write a program to compare two files character by character and check whether they are same or not?

Assignment Evaluation

0: Not Done [ ]	1: Incomplete [ ]	2: Late Submission [ ]
3: Needs Improvement [ ]	4: Complete [ ]	5: Well Done [ ]

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Assignment 11

Set A. Write C programs for the following problems.

1. Write a program to define value of pi and use it to find area and perimeter of a circle.
2. Write a program to display all command line arguments passed to main in the reverse order.
3. Write a program to accept three integers as command line arguments and find the minimum, maximum and average of the three. Display error message if invalid number of arguments are entered.
4. Define a macro MIN which gives the smallest number among 3 given numbers.

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Set B. Write C programs for the following problems.

1. Write a program to find square and cube of a number using macro – #define SQUARE(x), #define CUBE(x)
2. Write a program which accepts two integers as command line arguments. If second number is1 then display factorial of first number and if second number is 2 the check whether first number is even or odd. If the user enters invalid number of arguments, display appropriate message.
3. Write a program to accept two file names as command line argument and copy the contents of first file to second file.

Assignment Evaluation

0: Not Done [ ]	1: Incomplete [ ]	2: Late Submission [ ]
3: Needs Improvement [ ]	4: Complete [ ]	5: Well Done [ ]

Signature of Instructor **Date**